

Superinfection interference between dengue-2 and dengue-4 viruses in *Aedes aegypti* mosquitoes.

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Dengue virus consists of four antigenically distinct serotypes (DENV 1-4) that are transmitted to humans by *Aedes aegypti* and *Aedes albopictus* mosquitoes. In many dengue-endemic regions, co-circulation of two or more DENV serotypes is fairly common increasing the likelihood for exposure of the two vectors to multiple serotypes. We used a model system of DENV-2 and DENV-4 to investigate how prior exposure of *Aedes aegypti* to one DENV serotype affects its susceptibility to another serotype. *Aedes aegypti* mosquitoes were sequentially infected with DENV-2 and DENV-4 and the infection and dissemination rates for each virus determined. We found that prior infection of *Ae. aegypti* mosquitoes with DENV-4 rendered them significantly less susceptible to secondary infection with DENV-2. Although the results were not statistically significant, mosquitoes infected with DENV-2 were also less susceptible to secondary infection with DENV-4. The midgut dissemination and population dissemination rates for DENV-2 were significantly higher than those of DENV-4 when either virus was administered 7 days after administration of either a non-infectious blood meal or a blood meal containing a heterologous dengue serotype. These results demonstrate that superinfection interference between DENV serotypes is possible within *Ae. aegypti* mosquitoes, but its effect on DENV epidemiology may be dependent on the fitness of interacting serotypes.

Imported and autochthonous dengue in Spain.

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Dengue is globally the most important arboviral infection. It is caused by the dengue virus and it is generally transmitted by *Aedes* mosquitoes' bites (*Ae aegypti* or *Ae albopictus*). In Spain it was initially eradicated in the 20th century, together with the *Ae aegypti* vector, and currently most of the cases reported in Spain are imported by travelers from countries with dengue transmission (imported dengue). However, in recent years, cases of dengue have been described in people residing in Spain who had not traveled to areas with known transmission (autochthonous dengue), transmitted by *Aedes albopictus* (the so-called tiger mosquito), present especially in the Mediterranean basin. Therefore, a good knowledge of this potentially severe disease is required, so that it can be diagnosed early, and managed correctly, thus reducing its mortality, as well as its eventual autochthonous transmission.

Modulation of dengue virus infection of dendritic cells by *Aedes aegypti* saliva.

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Dengue virus (DV) is a flavivirus carried by the *Aedes aegypti* mosquito that causes a spectrum of illnesses in the tropics, including dengue fever, dengue hemorrhagic fever, and dengue shock syndrome. Dendritic cells (DCs) are professional antigen presenting cells recently shown to be permissive for DV, and implicated as the primary targets of initial DV infection. DV is transmitted to human host by infected mosquitoes during a blood meal, but it is currently unknown whether transmission is modified by vector saliva that is also deposited in the host's skin during feeding. Previous studies evaluated only the outcome of DV infection of DCs, and did not address the influence of mosquito saliva. To more fully

characterize natural transmission of DV, we evaluated the effects of *Ae. aegypti* saliva on DV infection of human myeloid DCs. We found that saliva inhibited DV infection in DCs. Moreover, pre-sensitization of DCs with saliva, prior to DV infection, enhanced inhibition. In addition, enhanced production of IL-12p70 and TNF-alpha were detected in DV-infected DC cultures exposed to mosquito saliva. The proportion of dead cells was also significantly reduced in these cultures. These data contribute to the overall understanding of the natural pathogenesis of DV infection and suggest that there is a protective role for mosquito saliva that limits viral uptake by DCs.

Mosquito densovirus significantly reduces the vector susceptibility to dengue virus serotype 2 in *Aedes albopictus* mosquitoes (Diptera: Culicidae).

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Dengue virus (DENV) is a major public health threat, with *Aedes albopictus* being the confirmed vector responsible for dengue epidemics in Guangzhou, China. Mosquito densoviruses (MDVs) are pathogenic mosquito-specific viruses, and a novel MDV was previously isolated from *Ae. albopictus* in Guangzhou. This study aims to determine the prevalence of MDVs in wild *Ae. albopictus* populations and investigate their potential interactions with DENV and impact on vector susceptibility for DENV. The prevalence of MDV in wild mosquitoes in China was investigated using open access sequencing data and PCR detection in *Ae. albopictus* in Guangzhou. The viral infection rate and titers in MDV-persistent C6/36 cells were evaluated at 12, 24, 48, 72, 96, and 120 h post infection (hpi) by indirect immunofluorescence assay (IFA) and real time quantitative PCR (RT-qPCR). The midgut infection rate (MIR), dissemination rate (DR), and salivary gland infection rate (SGIR) in various tissues of MDV-infected mosquitoes were detected and quantified at 0, 5, 10, and 15 days post infection (dpi) by RT-PCR and RT-qPCR. The chi-square test evaluated dengue virus serotype 2 (DENV-2) and *Aedes aegypti* densovirus (AaeDV) infection rates and related indices in mosquitoes, while Tukey's LSD and t-tests compared viral titers in C6/36 cells and tissues over time. The results revealed a relatively wide distribution of MDVs in *Aedes*, *Culex*, and *Anopheles* mosquitoes in China and an over 68% positive rate. In vitro, significant reductions in DENV-2 titers in supernatant at 120 hpi, and an apparent decrease in DENV-2-positive cells at 96 and 120 hpi were observed. In vivo, DENV-2 in the ovaries and salivary glands was first detected at 10 dpi in both monoinfected and superinfected *Ae. albopictus* females, while MDV superinfection with DENV-2 suppressed the salivary gland infection rate at 15 dpi. DENV-2 titer in the ovary and salivary glands of *Ae. albopictus* was reduced in superinfected mosquitoes at 15 dpi. MDVs is widespread in natural mosquito populations, and replication of DENV-2 is suppressed in MDV-infected *Ae. albopictus*, thus reducing vector susceptibility to DENV-2. Our study supports the hypothesis that MDVs may contribute to reducing transmission of DENV and provides an alternative strategy for mosquito-transmitted disease control.

[Dengue fever: clinical features].

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The vector for dengue fever and chikungunya, *Aedes albopictus*, was recently identified in Southeastern France, although the usual vector for dengue fever is *Aedes aegypti*, raising the possibility of cases occurring among the local population via viraemic individuals returning from endemic areas. Dengue fever is usually transmitted by *Aedes aegypti*. It is due to an arbovirus-flavivirus of which four different

serotypes are known: Den 1 to 4. Each serotype is responsible for specific prolonged immunity but no cross-reactivity exists between serotypes. Clinically, the onset is abrupt with frontal headache, retro-orbital pain, myalgia, joint pain, prostration and, in many cases, a macular rash usually sparing the face and extremities. Haemorrhagic signs may occur, such as petechiae, purpura, epistaxis or bleeding gingivae. Two severe forms of dengue fever, particularly among children below 3 years of age, include dengue haemorrhagic fever (DHF) and DHF with shock (dengue shock syndrome). If a case is suspected in metropolitan France, the diagnosis should be systematically confirmed by positive specific IgM, RT-PCR or viral isolation. Treatment of dengue fever, whether in its uncomplicated form or with hemorrhagic manifestations or shock, remains symptomatic. There is no specific anti-viral treatment. A case should be notified to allow French health authorities to take the appropriate measures for vector control.

Environmental influences on *Aedes aegypti* (Diptera: Culicidae) population densities across 2 urban communities on the U.S.-Mexico border.

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The mosquito species *Aedes aegypti* (Linnaeus) is the vector of multiple arboviruses, including dengue, Chikungunya, Zika, and yellow fever. Risk of infections associated with these arboviruses continues to expand as the geographical range of *Ae. aegypti* extends into temperate regions. Although *Ae. aegypti* is abundant along the U.S.-Mexico border, the ecology of this mosquito species in this temperate/subtropical desert is not well understood. Our study objective was to estimate the seasonal population density in 2 urban communities: Sparks, El Paso, Texas and Anapra, Ciudad Juárez, Chihuahua, Mexico. Estimates of the population density of adult *Ae. aegypti* were obtained by month (June to December) and across years (2016 to 2018) using gravid traps. Mosquitoes were collected inside and outside a total of 108 and 101 participating homes in Sparks and Anapra, respectively. We determined multiple environmental and climatic factors influencing annual population trends. Generally, an increase in the abundance of *Ae. aegypti* was associated with an increase in precipitation, moderate temperatures, and high humidity, while months with temperatures below 4.4°C led to near absence of adults. Across months, we found low densities of *Ae. aegypti* during June and July, followed by near 100-fold increases in abundance from August to October before rapidly decreasing to near zero in subsequent cold season months. Our study sheds light on population dynamics and response variables at the leading edge of *Ae. aegypti* range, which require the development of targeted vector control measures for this mosquito species in this and other regions.

Risk factors for the presence of chikungunya and dengue vectors (*Aedes aegypti* and *Aedes albopictus*), their altitudinal distribution and climatic determinants of their abundance in central Nepal.

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The presence of the recently introduced primary dengue virus vector mosquito *Aedes aegypti* in Nepal, in association with the likely indigenous secondary vector *Aedes albopictus*, raises public health concerns. Chikungunya fever cases have also been reported in Nepal, and the virus causing this disease is also transmitted by these mosquito species. Here we report the results of a study on the risk factors for the presence of chikungunya and dengue virus vectors, their elevational ceiling of distribution, and climatic

determinants of their abundance in central Nepal. We collected immature stages of mosquitoes during six monthly cross-sectional surveys covering six administrative districts along an altitudinal transect in central Nepal that extended from Birgunj (80 m above sea level [asl]) to Dhunche (highest altitude sampled: 2,100 m asl). The dengue vectors *Ae. aegypti* and *Ae. albopictus* were commonly found up to 1,350 m asl in Kathmandu valley and were present but rarely found from 1,750 to 2,100 m asl in Dhunche. The lymphatic filariasis vector *Culex quinquefasciatus* was commonly found throughout the study transect. Physiographic region, month of collection, collection station and container type were significant predictors of the occurrence and co-occurrence of *Ae. aegypti* and *Ae. albopictus*. The climatic variables rainfall, temperature, and relative humidity were significant predictors of chikungunya and dengue virus vectors abundance. We conclude that chikungunya and dengue virus vectors have already established their populations up to the High Mountain region of Nepal and that this may be attributed to the environmental and climate change that has been observed over the decades in Nepal. The rapid expansion of the distribution of these important disease vectors in the High Mountain region, previously considered to be non-endemic for dengue and chikungunya fever, calls for urgent actions to protect the health of local people and tourists travelling in the central Himalayas.

Inhibition and Larvicidal Activity of Phenylpropanoids from *Piper sarmentosum* on Acetylcholinesterase against Mosquito Vectors and Their Binding Mode of Interaction.

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Aedes aegypti, *Aedes albopictus* and *Culex quinquefasciatus* are vectors of dengue fever and West Nile virus diseases. This study was conducted to determine the toxicity, mechanism of action and the binding interaction of three active phenylpropanoids from *Piper sarmentosum* (Piperaceae) toward late 3rd or early 4th larvae of above vectors. A bioassay guided-fractionation on the hexane extract from the roots of *Piper sarmentosum* led to the isolation and identification of three active phenylpropanoids; asaricin 1, isoasarone 2 and trans-asarone 3. The current study involved evaluation of the toxicity and acetylcholinesterase (AChE) inhibition of these compounds against *Aedes aegypti*, *Aedes albopictus* and *Culex quinquefasciatus* larvae. Asaricin 1 and isoasarone 2 were highly potent against *Aedes aegypti*, *Aedes albopictus* and *Culex quinquefasciatus* larvae causing up to 100% mortality at $\leq 15 \mu\text{g/mL}$ concentration. The ovicidal activity of asaricin 1, isoasarone 2 and trans-asarone 3 were evaluated through egg hatching. Asaricin 1 and isoasarone 2 showed potent ovicidal activity. Ovicidal activity for both compounds was up to 95% at $25\mu\text{g/mL}$. Asaricin 1 and isoasarone 2 showed strong inhibition on acetylcholinesterase with relative IC₅₀ values of 0.73 to 1.87 $\mu\text{g/mL}$ respectively. These findings coupled with the high AChE inhibition may suggest that asaricin 1 and isoasarone 2 are neuron toxic compounds toward *Aedes aegypti*, *Aedes albopictus* and *Culex quinquefasciatus*. Further computational docking with Autodock Vina elaborates the possible interaction of asaricin 1 and isoasarone 2 with three possible binding sites of AChE which includes catalytic triads (CAS: S238, E367, H480), the peripheral sites (PAS: E72, W271) and anionic binding site (W83). The binding affinity of asaricin 1 and isoasarone 2 were relatively strong with asaricin 1 showed a higher binding affinity in the anionic pocket.

Multiplication of chikungunya virus in salivary glands of *Aedes albopictus* (Oahu strain) mosquitoes: an electron microscopic study.

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Aedes albopictus as well as *Aedes aegypti* is an important vector of chikungunya and dengue viruses. Electron microscopic observations on the salivary glands of *Ae. albopictus* infected with chikungunya virus were performed in comparing with those of *Ae. aegypti* infected with dengue virus. No virus budding from the cell surface of the chikungunya-infected mosquito's salivary glands was found as shown in dengue-infected ones, in contrast to the findings of the mammalian cells such as Vero, KB, IMR, J-111 and BHK-21 cells infected with chikungunya and/or dengue virus(es).

[An etiological and serological study on Dengue epidemic in Guangdong Province].

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In 1978-1995, Dengue fever epidemics (serotype 1-4) occurred in Guangdong Province. Among 9 species of mosquitoes found in the epidemic areas, the positive rate of virus isolation in *Aedes albopictus* was 20.69%, in *Aedes aegypti* 18.95%, in *Culex quinquefasciatus* 10.29%, Dengue viruses were not isolated in other 6 species of mosquitoes. The positive rate of virus isolation in the three species showed no significant differences ($\chi^2 = 2.77$ $P > 0.05$). *Aedes aegypti* was the main transmitting vector in Hainan Island and Zhanjiang District, whereas *Aedes albopictus* was in the other epidemic areas. The positive rate of virus isolation within 3 days of disease onset was between 73.58% and 82.05%, but the virus could still be isolated on the 8th day of disease onset. The highest antibody positive rate from the patients' paired-sera was 93.89% and the lowest was 33.71%. IgM antibody could be detected 5 days after the disease onset in some cases, even on the 33rd day of disease onset. These results suggest that virus isolation by collecting patient's blood within 1-4 days of disease onset, or IgM antibody detection after 5 days of disease onset can be used as method for early and rapid diagnosis of the disease.
